INCH-POUND

MIL-S-52506D(ME)
30 July 1992
SUPERSEDING
MIL-S-52506C(ME)
27 December 1984

MILITARY SPECIFICATION

# SEALS, SHAFT, MECHANICAL, LOW-PRESSURE TYPE, SELF-LUBRICATING: FOR GENERAL-PURPOSE CENTRIFUGAL PUMPS

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers four types and 16 sizes of low-pressure, self-lubricating, mechanical shaft seals, commonly used on centrifugal pumps.

1.2 Classification. The seals shall be of the following types and sizes, as specified (see 6.2):

Type I - Pressed-in type with stationary sealing washer and rotating seating surface (see figure 1).

Size 3 - for 0.375-inch diameter shafts.
Size 4 - for 0.500-inch diameter shafts.
Size 5 - for 0.625-inch diameter shafts.
Size 6 - for 0.750-inch diameter shafts.

Type II - Short-length type with rotating sealing washer and stationary seating surface (see figure 2).

Size 3 - for 0.375-inch diameter shafts.
Size 4 - for 0.500-inch diameter shafts.
Size 5 - for 0.625-inch diameter shafts.
Size 6 - for 0.750-inch diameter shafts.
Size 7 - for 0.875-inch diameter shafts.
Size 8 - for 1.000-inch diameter shafts.
Size 8-1/2 - for 1.062-inch diameter shafts.
Size 9 - for 1.125-inch diameter shafts.
Size 10 - for 1.250-inch diameter shafts.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in Improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type III - Small-diameter type with rotating sealing washer and stationary seating surface (see figure 3).

Size 3 - for 0.375-inch diameter shafts.
Size 4 - for 0.500-inch diameter shafts.
Size 5 - for 0.625-inch diameter shafts.
Size 6 - for 0.750-inch diameter shafts.
Size 7 - for 0.875-inch diameter shafts.
Size 8 - for 1.000-inch diameter shafts.
Size 9 - for 1.125-inch diameter shafts.
Size 10 - for 1.250-inch diameter shafts.

Type IV - Flexible-for-misalignment type with rotating sealing washer and stationary seating surface (see figure 4).

Size 8 - for 1.000-inch diameter shafts.
Size 8-1/2 - for 1.062-inch diameter shafts.
Size 9 - for 1.125-inch diameter shafts.
Size 10 - for 1.250-inch diameter shafts.
Size 12 - for 1.500-inch diameter shafts.
Size 14 - for 1.750-inch diameter shafts.
Size 16 - for 2.000-inch diameter shafts.
Size 18 - for 2.250-inch diameter shafts.
Size 20 - for 2.500-inch diameter shafts.
Size 22 - for 2.750-inch diameter shafts.
Size 24 - for 3.000-inch diameter shafts.

## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Hood Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-I-10 - Insulating Materials, Electrical, Ceramic, Class L.
MIL-P-116 - Preservation, Methods of.



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STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-889 | - Dissimilar Metals. |

(Unless otherwise indicated, copies of federal and military specifications, and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| A 167 | - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip. |
| A 313 | - Chromium-Nickel Stainless and Heat-Resisting Steel Spring Wire. |
| A 319 | - Gray Iron Castings for Elevated Temperatures for Non-Pressure Containing Parts. |
| C 135 | - True Specific Gravity of Refractory Materials by Water Immersion. |
| C 086 | - Scleroscope Hardness Testing of Fine-Grain Carbon and Graphite Materials. |
| D 2000 | - Elastomeric Materials for Automotive Applications. |
| D 3953 | - Strapping, Flat Steel and Seals. |
| D 4675 | - Selection and Use of Flat Strapping Materials. |
| E 8 | - Tension Testing of Metallic Materials. |
| E 9 | - Compression Testing of Metallic Materials at Room Temperature. |
| E 228 | - Linear Thermal Expansion of Solid Materials with a Vitreous Silica Dilatometer. |

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. (NMFTA)

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)



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UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, ATTN : Tariff Publishing Officer, Room 1106, 222 S. Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Each seal shall consist essentially of a seat, seat ring, carbon seal washer, bellows, retainer, spring, drive band, and spring holder.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Cadmium plating. Cadmium plating and cadmium plated items shall not be used.

3.3.2 Material deterioration prevention and control. The seal shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environments to which the seal may be exposed.

3.3.2.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.2.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the seals may be newly fabricated from recovered materials to the maximum extent practicable, provided the seals produced meets all other requirements of this specification.



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Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the seals.

3.4 Sealing washer. The sealing washer shall be a mixture of carbon, graphite, and either babbitt or resin. The sealing washer shall have the following physical properties:

| | |
|---|---|
| Hardness, minimum | 60 (using ASTM C 886) |
| Compressive strength, minimum | 32,000 pounds per square inch (psi) |
| Transverse strength, minimum | 10,000 psi |
| Tensile strength, minimum | 2900 psi |
| Specific gravity | 1.75 minimum - 3.20 maximum |
| Coefficient of linear thermal expansion, maximum | 2.95 X $10^{-6}$ (1/°F) |
| Resistance to heat as specified (see 6.2): | |
| Resin type | 500 °F (260 °C) |
| Babbitt type | 300 °F (149 °C) |

3.5 Seat. The seat shall be gray iron alloy conforming to ASTM A 319, class I or II; or ceramic material conforming to MIL-I-10, grade L524 C, as specified (see 6.2).

3.6 Static seals. The seat ring or the O-ring and the bellows shall be an elastomer meeting the requirements of ASTM D 2000, line call out M2BG614A14B14EF21034F17. All rubber seals shall be less than 12 months old on date of acceptance by the Government.

3.7 Retainer. The retainer shall be corrosion-resistant steel conforming to ASTM A 167, UNS S30400.

3.8 Spring. The spring shall be corrosion-resistant steel conforming to ASTM A 313, UNS S30200 or S30400.

3.9 Drive band and spring holder. The drive band and spring holder shall be corrosion-resistant steel conforming to ASTM A 167, UNS S30400.

3.10 Seal contact surfaces. The seal contact surfaces shall be lapped flat and smooth to not more than 4 light bands for the softer surface and not more than 3 light bands for the harder surface.

3.11 Performance. The seal shall prevent leakage of liquid between the rotating pump shaft-and the adjacent stationary pump parts at the shaft speed and pressure specified on the applicable figures. There shall be no accumulation of escaped liquid from the seal after run-in.

3.12 Lubrication. The seal shall be lubricated with light oil during initial installation of the seal. Subsequently, the seal shall be continuously self-lubricating utilizing the fluid being pumped.

3.13 Identification marking. The seal shall be identified in accordance with MIL-STD-130 and shall include part or identifying numbers (see 6.4).



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3.14 Workmanship. All parts, components, and assemblies of the seal including casting, molded parts, stampings, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All edges shall be rounded or beveled.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all Inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article Inspection.

4.3.1 Examination. The first article seal shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article seal shall be tested as specified in 4.5.2.1 through 4.5.2.5. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105 table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.



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4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.2, 4.5.2.3, and 4.5.2.4, except that the run-in time shall be the contractor's option. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The seal shall be examined as specified herein for the following major defects:

Major

101. Type or size not as specified (see 1.2).
102. Assembly incorrect (see 3.1).
103. Cadmium plating or plated items used (see 3.3.1).
104. Materials are not resistant to corrosion and deterioration, or not treated to be resistant to corrosion and deterioration for the applicable storage and operating environment (see 3.3.2).
105. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2.1).
106. Contractor does not have documentation available for identification of material, material finishes or treatment (see 3.3.2.2).
107. Used, rebuilt or remanufactured components, pieces or parts incorporated in the seal (see 3.3.2.3),
108. Sealing washer not as specified (see 3.4).
109. Seat not as specified (see 3.5).
110. Static seals not as specified (see 3.6).
111. Retainer not as specified (see 3.7).
112. Spring not as specified (see 3.8).
113. Drive band and spring holder not as specified (see 3.9).
114. Seal contact surfaces not as specified (see 3.10).
115. Lubrication not as specified (see 3.12).
116. Identification marking missing, incomplete, or illegible (see 3.13).
117. Workmanship not as specified (see 3.14).

4.5.2 Tests.

4.5.2.1 Carbon sealing washer physical tests. The test specimens used for the physical tests of the sealing washer shall be made from a completed sealing washer or a specimen of a different shape made of the same material as that in the carbon washers.

4.5.2.1.1 Scleroscope hardness. The hardness test shall be conducted in accordance with ASTM C 886. An average hardness reading of less than 60 shall constitute failure of this test.

4.5.2.1.2 Compressive strength. The compressive strength test shall be conducted in accordance with ASTM E 9. A compressive strength of less than 32,000 psi shall constitute failure of this test.



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4.5.2.1.3 Transverse strength. A test specimen 2,50 inches long by 0.50 inch square shall be used for this test. The test specimen shall be supported on two knife edged rounded on a radius of 0.062 inch. The knife edges shall be 2.0 inches apart and shall be equidistant from the specimen ends. The stress shall be applied by a third knife edge, midway between the support knife edges and in the direction of molding pressure. The force shall be applied at a steady rate not exceeding 2 pounds per second. The total net force in pounds at which the specimen ruptures shall be noted and recorded. Transverse strength shall then be calculated in accordance with the following formula:

$$\text{Transverse strength in pounds per square inch} = \frac{3LF}{2WT2}$$

where: F = total force in pounds at the instant of rupture.
w = width of specimen in inches.
T = thickness of specimen in inches.
L = distance between supporting knife edges.

A transverse strength of less than 10,000 psi shall constitute failure of this test.

4.5.2.1.4 Tensile strength. Tensile strength shall be determined in accordance with ASTM E 8. A tensile strength less than 2900 psi shall constitute failure of this test.

4.5.2.1.5 Specific gravity. Specific gravity shall be determined in accordance with ASTM C 135. A specific gravity of less than 1.75 or more than 3.20 shall constitute failure of this test.

4.5.2.1.6 Coefficient of linear thermal expansion. The coefficient of linear thermal expansion shall be determined in accordance with ASTM E 228. A coefficient of linear thermal expansion greater than 2.95 X $10^{-6}$ (1/°F) shall constitute failure of this test..

4.5.2.1.7 High-temperature test. One carbon sealing washer shall be placed in a circulating-hot-air oven and maintained for 4 hours at 300 °F (149 °C) for washers containing babbitt and at 500 °F (260 °C) for washers containing resin. The washer shall then be removed from the oven and allowed to cool in air until it returns to room temperature. The washer shall be examined for evidence of cracks, spalling, or surface deterioration. Any evidence of cracks, spalling, or surface deterioration shall constitute failure of this test.

4.5.2.2 Flatness. The flatness of the sealing surfaces of a carbon sealing washer and its corresponding seat shall be determined under a monochromatic light using an optical flat. The monochromatic light shall be transmitted as a strong near-one-color light of 11.6 millionths of an inch per dark interference band. The optical flat shall be highly-transparent, abrasion-resistant quartz with good wear characteristics and thermal stability. The optical flat shall be produced to a standard of 0.10 of a light band (0.00000116-inch) and 0.20 of a light band (0.00000232-inch) across the diameter. A flatness not within 4 light bands for the softer surface or a flatness not within 3 light bands for the harder surface shall constitute failure of this test.



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4.5.2.3 Run-in test. The seal assembly shall be installed in a pump or in a fixture that will dynamically simulate pump operating conditions. Water shall be used as the test fluid. The pump or fixture shall be operated for 100 hours continuously at the speed and pressure indicated in the applicable figure and stopped. All the area on the fixture or pump which would collect moisture escaping past the seal shall be wiped clean and dry. Operate the pump or fixture at the same conditions for 20 more hours. Any accumulation of liquid or appearance of a wet area caused by liquid escaping past the seal shall constitute failure of this test.

4.5.2.4 Hydrostatic test. The seal, after successfully completing the run-in test (see 4.5.2.3), shall be subjected to 1.5 times the operating pressure indicated in the applicable figure for a period of 5 minutes. During this test the pump or fixture shall not be operated. Leakage through the seal shall constitute failure of this test.

4.5.2.5 Static seals. The static seals shall be testing in accordance with ASTM D 2000, line call out M2BG614A14B14EF21034F17. Failure to conform to any of the applicable requirements of ASTM D 2000 shall constitute failure of this test.

4.6 Inspection of packaging

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

118. Preservation not as specified for level A, B and C (see 5.1.1, 5.1.2 and 5.1.3).
119. Consolidation not as specified for level A, B and C (see 5.1.1, 5.1.2 and 5.1.3).
120. Boxes for level A and B packing not of type specified (see 5.2.1 and 5.2.2).
121. Gross weight of shipping containers not as specified for level A and B (see 5.2.1 and 5.2.2).
122. Shipping containers not strapped as specified for level A and B (see 5.2.1 and 5.2.2).
123. Marking Illegible, incorrect, incomplete, or missing (see 5.3).
124. Packing for level C not as specified by the referenced documents (see 5.2.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).



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5.1.1 Level A. Each seal shall be preserved in accordance with MIL-P-116. method IC-2, using a close-fitting box conforming to PPP-B-566, style, type, and class optional, as the container.

5.1.1.1 Consolidation. Seals of like description preserved as specified in 5.1.1 shall be consolidated in quantities of 10 using a close-fitting box conforming to PPP-B-636, W6c, style optional, as the container.

5.1.2 Level B. Each seal shall be preserved in accordance with MIL-P-116, method III using a close-fitting box conforming to PPP-B-566, style, type, and class optional, as the container.

5.1.2.1 Consolidation. Seals of like description preserved as specified in 5.1.2 shall be consolidated in quantities of 10 using a close-fitting box conforming to PPP-B-636, W6c, style optional, as the container.

5.1.3 Level C. The seals shall be preserved as specified for level B except the consolidation containers shall be domestic class.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The seals, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. Strapping shall conform to ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675. Gross weight of shipping containers shall not exceed 100 pounds.

5.2.2 Level B. Seals, preserved as specified in 5.1, shall be packed in boxes conforming to PPP-B-636, grade V3c, style optional. Weight of contents in shipping containers shall not exceed the limit stated in the box specification. Flat steel strapping shall be applied in accordance with the box specification.

5.2.3 Level C. Seals, preserved as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature which is helpful, but is not mandatory.)

6.1 Intended use. The seals are intended to be used as a component of centrifugal pumps for sealing potable water and hydrocarbons including such liquids as gasoline, diesel fuel, and jet fuel.



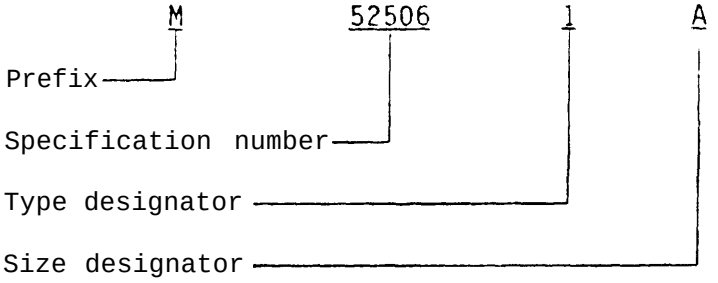
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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Type and size of seal required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Time frame required for submission of first article model and number of seals required (see 3.2).
e. Whether heat-resistant capability of the sealing washer shall be 300 °F (149 °C) or 500 °F (260 °C) (see 3.4).
f. Whether ceramic or gray iron alloy seat is required (see 3.5).
g. Level of preservation and level of packing required (see 5.1 and 5.2).
h. Any special markings (see 5.3).
i. Part or identifying number required (see 3.13 and 6.4).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Part or identifying number (PIN). The PIN to be used for seals acquired to this specification is designed as follows:






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6.4.1 Type designator. A one position field used to designate the required type (see table I).

TABLE I. Type.

| Number designator | Type designator |
|---|---|
| 1 | I - Pressed-in type with stationary sealing washer and rotating seating surface |
| 2 | II - Short-length type with rotating sealing washer and stationary seating surface |
| 3 | III - Small-diameter type with rotating sealing washer and stationary seating surface |
| 4 | IV - Flexible-for-misalignment type with rotating sealing washer and stationary seating surface |

6.4.2 Size designator. A one position field used designate the required size (see table II).

TABLE II. Size.

| Letter designator | Size designator |
|---|---|
| A | Size 3 - for 0.375-inch diameter shafts |
| B | Size 4 - for 0.500-inch diameter shafts |
| C | Size 5 - for 0.625-inch diameter shafts |
| D | Size 6 - for 0.750-inch diameter shafts |
| E | Size 7 - for 0.875-inch diameter shafts |
| F | Size 8 - for 1.000-inch diameter shafts |
| G | Size 8-1/2 - for 1.062-inch diameter shafts |
| H | Size 9 - for 1.125-inch diameter shafts |
| I | Size 10 - for 1.250-inch diameter shafts |

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



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6.6 Subject term (key word) listing.

Seal, low pressure
Seal, mechanical
Seal, self lubricating

Custodian:
Army - ME

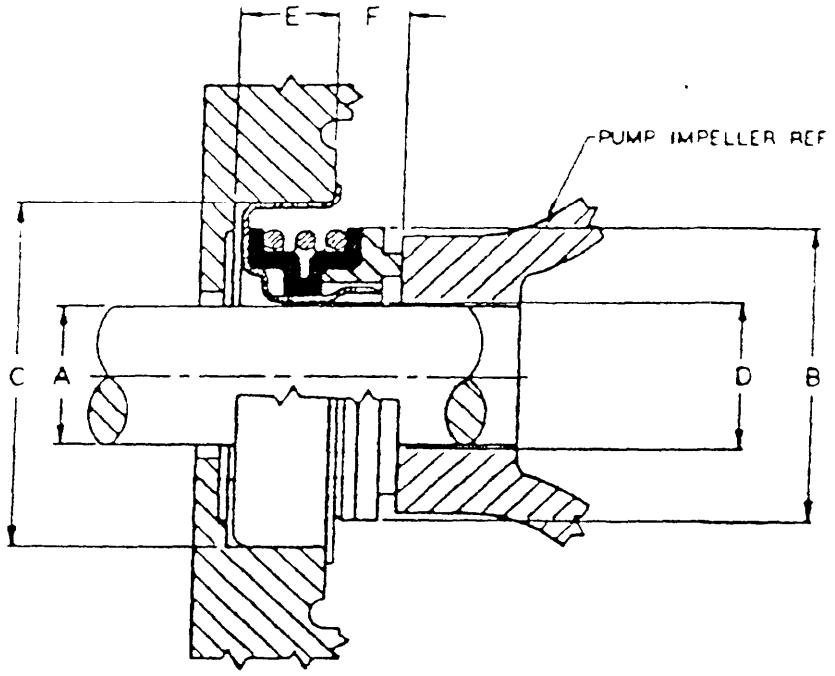
Review activity:
DLA - CS

Preparing activi
Army - ME

Project 4320-A34



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| SIZE | A SHAFT DIA | B SEALING WASH. OD | C SEAT BORE | D SEAL ID | E CAVITY DEPTH | F WORKING LENGTH | OPER PRESS. PSI | SHAFT SPEED RPM | APPLICATION |
|---|---|---|---|---|---|---|---|---|---|
| 3 | .375 | — | 1.125 | 13/32 | .343 | 1/16 | 75 | 7500 | AUTOMOTIVE WATER |
| 4 | .500 | 1 1/16 | 1.125 | 17/32 | .350 | 3/16 | 75 | 7500 | PUMPS, SHALLOW WELL |
| 5 | .625 | 1 9/32 | 1.435 | 45/64 | .350 | 17/64 | 75 | 7500 | JET PUMPS, SMALL HIGH |
| 6 | .750 | 1 17/16 | 1.575 | 13/16 | .380 | 15/64 | 75 | 7500 | SPEED CENT PUMPS, ETC. |

NOTES:

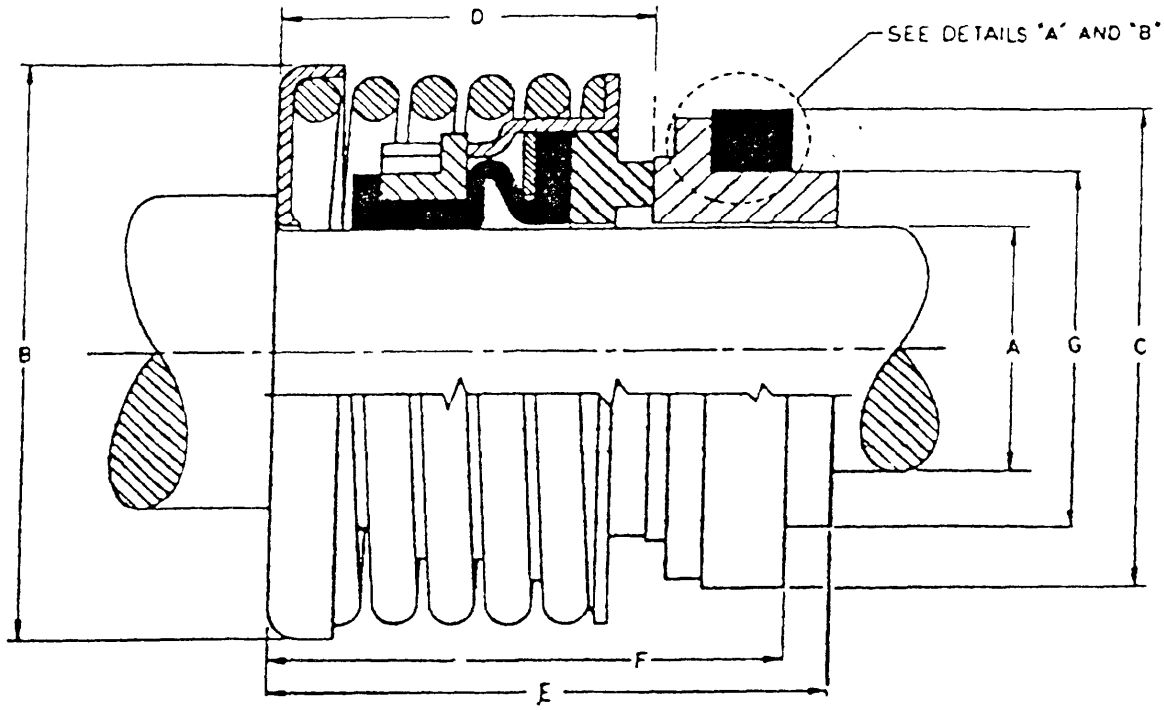
1. DIMENSIONS SHOWN ARE IN INCHES.
2. TOLERANCES:
   - A. DECIMAL, INCHES —— ±.005
   - B. FRACTIONAL, INCHES —— ± 1/64
   - C. PRESSURE, PSI —— ± 8
   - D. SPEED, RPM —— ± 50

FIGURE 1. Seal, shaft, pressed, Type I.

X-274B



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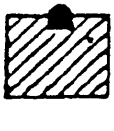



| SIZE | A SHAFT DIA | B SEAL OD (MAX) | C SEAT BORE | D WORK LENGTH | E OVER LENGTH | F OVER LENGTH | G SEAT TAIL BORE | OPER PRESS PSI | SHAFT SPEED RPM | APPLICATION |
|---|---|---|---|---|---|---|---|---|---|---|
| 3 | .375 | 1-1/32 | .875 | .812 | 1.187 | 1.125 | 0.625 | 200 | 3600 | GENERAL PURPOSE SEALS FOR WIDE VARIETY OF PUMPING APPLICATIONS WHERE LENGTH MUST BE KEPT TO A MIN AND OD IS NOT A LIMITING FACTOR |
| 4 | .500 | 1-3/16 | 1.000 | .812 | 1.187 | 1.125 | 0.750 | 200 | 3600 | |
| 5 | .625 | 1-3/8 | 1.250 | .875 | 1.343 | 1.281 | 0.937 | 200 | 3600 | |
| 6 | .750 | 1-1/2 | 1.375 | .875 | 1.343 | 1.281 | 1.062 | 200 | 3600 | |
| 7 | .875 | 1-5/8 | 1.500 | .937 | 1.406 | 1.437 | 1.187 | 200 | 3600 | |
| 8 | 1.000 | 1-31/32 | 1.625 | .937 | 1.500 | 1.437 | 1.312 | 200 | 3600 | |
| 8-1/2 | 1.062 | 2-1/32 | 1.687 | .937 | 1.562 | 1.437 | 1.375 | 200 | 3600 | |
| 9 | 1.125 | 2-1/8 | 1.750 | 1.062 | 1.562 | 1.562 | 1.437 | 200 | 3600 | |
| 10 | 1.250 | 2-1/4 | 1.875 | 1.062 | 1.562 | 1.562 | 1.562 | 200 | 3600 | |

NOTES:
1. DIMENSIONS SHOWN ARE IN INCHES.
2. TOLERANCES:
   - A. DECIMAL, INCHES ——— ±.005
   - B. FRACTIONAL, INCHES ——— ± 1/64
   - C. PRESSURE, PSI ——— + 0, − 5
   - D. SPEED, RPM ——— ± 50
3. COL. E NOT APP. TO DETAIL "A" OR "B".
4. COL. G NOT APP. TO DETAIL "A" OR "B".



DETAIL "A" ALTERNATE SEAT SEAL, CUP



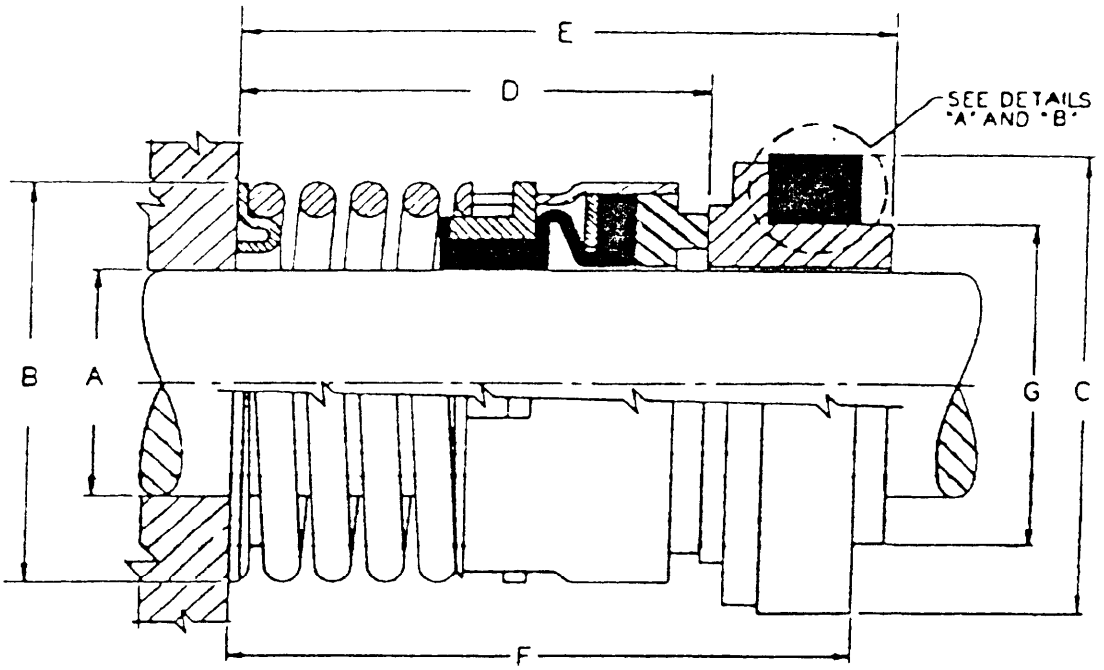
DETAIL "B" ALTERNATE SEAT SEAL, "O" RING

FIGURE 2. Seal, shaft, short length, Type II.

X-275B



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| SIZE | A SHAFT DIA | B SEAL OD | C SEAT BORE | D WORK LENGTH | E OVER LENGTH | F OVER LENGTH | G SEAT TAIL BORE | OPER PRESS PSI | SHAFT SPEED RPM | APPLICATION |
|---|---|---|---|---|---|---|---|---|---|---|
| 3 | 0.375 | 15/16 | 0.875 | 1.187 | 1.562 | 1.500 | 0.625 | 200 | 3600 | SAME AS TYPE II-EXCEPT USED WHERE OD MUST BE KEPT TO A MIN AND WHERE LENGTH IS NOT A LIMITING FACTOR |
| 4 | 0.500 | 15/16 | 1.000 | 1.187 | 1.562 | 1.500 | 0.750 | 200 | 3600 | |
| 5 | 0.625 | 1-3/32 | 1.250 | 1.312 | 1.781 | 1.719 | 0.937 | 200 | 3600 | |
| 6 | 0.750 | 1-3/32 | 1.375 | 1.312 | 1.781 | 1.719 | 1.062 | 200 | 3600 | |
| 7 | 0.875 | 1-3/32 | 1.500 | 1.375 | 1.843 | 1.781 | 1.187 | 200 | 3600 | |
| 8 | 1.000 | 1-1/2 | 1.625 | 1.562 | 2.062 | 2.062 | 1.312 | 200 | 3600 | |
| 9 | 1.125 | 1-5/8 | 1.750 | 1.625 | 2.125 | 2.125 | 1.437 | 200 | 3600 | |
| 10 | 1.250 | 1-3/16 | 1.875 | 1.625 | 2.125 | 2.125 | 1.562 | 200 | 3600 | |

NOTES:
1. DIMENSIONS SHOWN ARE IN INCHES.
2. TOLERANCES:
   - A. DECIMAL, INCHES ——— ±.005
   - B. FRACTIONAL, INCHES ——— ±1/64
   - C. PRESSURE, PSI ——— +0 −5
   - D. SPEED, RPM ——— ±50
3. COL. E NOT APP. TO DETAIL "A" OR "B".
4. COL. G NOT APP. TO DETAIL "A" OR "B".



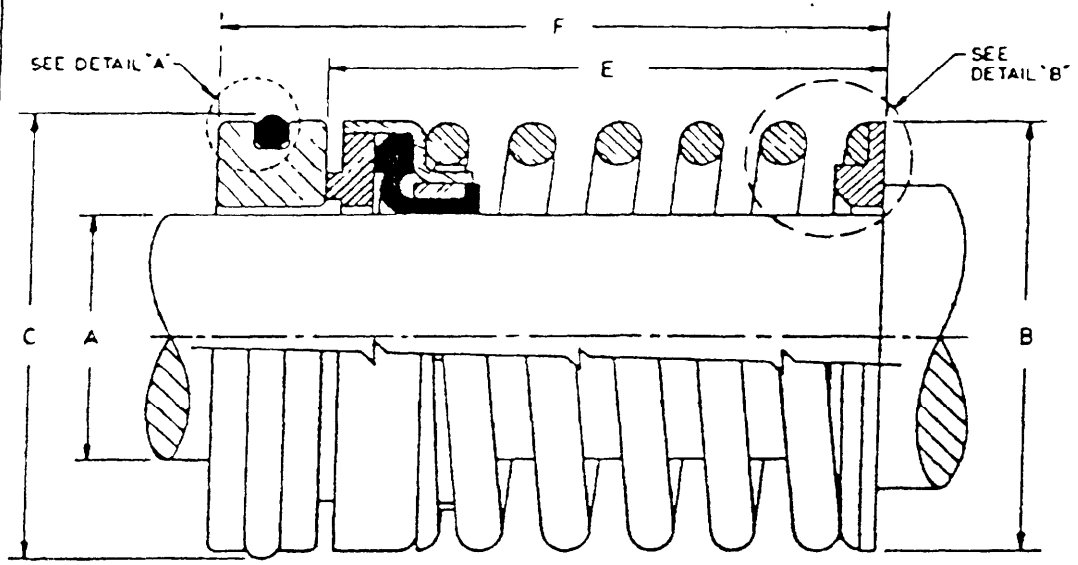
DETAIL "A" ALTERNATE SEAT SEAL, CUP

DETAIL "B" ALTERNATE SEAT SEAL, "O" RING

FIGURE 3. Seal, shaft, small diameter, Type III.



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| SIZE | A SHAFT DIA | B SEAL OD | C SEAT BORE | D SEAT ID | E WORK LENGTH | F OVER LENGTH | OPER PRESS PSI | SHAFT SPEED RPM | APPLICATION |
|---|---|---|---|---|---|---|---|---|---|
| 8 | 1.000 | 1-11/16 | 1.625 | 1-3/32 | 1.000 | 1.437 | 200 | 3600 | CENTRIFUGAL, ROTARY, TURBINE PUMPS WHERE ECONOMY IS A MAJOR FACTOR |
| 8-1/2 | 1.062 | 1-11/16 | 1.750 | 1-7/64 | 1.000 | 1.375 | 200 | 3600 | |
| 9 | 1.125 | 1-13/16 | 1.750 | 1-3/16 | 1.062 | 1.499 | 200 | 3600 | |
| 10 | 1.250 | 1-15/16 | 1.875 | 1-5/16 | 1.062 | 1.499 | 200 | 3600 | |
| 12 | 1.500 | 2-3/16 | 2.125 | 1-5/8 | 1.125 | 1.562 | 200 | 3600 | |
| 14 | 1.750 | 2-5/8 | 2.500 | | 1.375 | 1.875 | 200 | 3600 | |
| 16 | 2.000 | 2-7/8 | 2.750 | | 1.500 | 2.000 | 200 | 3600 | |
| 18 | 2.250 | 3-1/8 | 3.125 | | 1.687 | 2.250 | 200 | 3600 | |
| 20 | 2.500 | 3-11/32 | 3.375 | | 1.812 | 2.375 | 200 | 3600 | |
| 22 | 2.750 | 3-3/4 | 3.500 | | 1.937 | 2.562 | 200 | 3600 | |
| 24 | 3.000 | 4 | 3.875 | | 2.062 | 2.687 | 200 | 3600 | |

NOTES:
1. DIMENSIONS SHOWN ARE IN INCHES.
2. TOLERANCES:
   A. DECIMAL, INCHES ——— ±.005
   B. FRACTIONAL, INCHES — ± 1/64
   C. PRESSURE, PSI ——— +0 -5
   D. SPEED, RPM ——— ±50


DETAIL "A"
ALTERNATE
SEAT SEAL


DETAIL "B"
ALTERNATE
SPRING HOLDER

FIGURE 4. Seal, shaft, flexible, Type IV.



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3. DOCUMENT TITLE Seals, Shaft, Mechanical, Low-Pressure Type, Self-Lubricating: for General-Purpose Centrifugal Pumps

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